

END SEMESTER ASSESSMENT (ESA) - Jan - May 2024**UE21CS343BB7 - Mobile and Autonomous Robotics****Total Marks : 100.0**

1.a. Discuss the importance of technical, social, and ethical challenges in the development of autonomous Robots, with specific examples for each challenge.
(7.0 Marks)

1.b. What is forward and inverse kinematics in robotics? Why is inverse kinematics important, and how does it work in the context of robot arm manipulators?
(7.0 Marks)

1.c. Explain the role and functionality of encoders in robotics. What factors should be considered when selecting an encoder for a specific application in robotics.
(7.0 Marks)

1.d. Distinguish between active and passive sensors, and give examples of each type. (4.0 Marks)

2.a. What is the difference between locomotion and manipulation? Discuss the key issues and limitations affecting locomotion in robotic systems. (7.0 Marks)

2.b. What is static and dynamic stability in a robot? Explain support polygon with example. (7.0 Marks)

2.c. Explain the concept of degrees of freedom (DOF) in robotics and describe the capabilities of robots with different DOF levels, specifically 2-DOF, 3-DOF, 5-DOF, and 6-DOF. Provide examples of each type of robot and discuss their potential applications. (7.0 Marks)

2.d. Write four differences between legged and wheeled robots.

(4.0 Marks)

3.a. What do optical encoders measure? How do they work? How does the sensor on the opposite side of the light source react when the encoder disk blocks the light beam?

(7.0 Marks)

3.b. What is corner detection in feature extraction? Explain Moravec's corner detection algorithm with the help of a diagram. What is the improvement introduced by the Harris corner detector?

(7.0 Marks)

3.c. Define the IMU and specify the number of degrees of freedom it typically estimates. Provide a schematic representation of its block diagram.

(7.0 Marks)

3.d. What is dead reckoning? Which sensor is used for dead reckoning? (4.0 Marks)

4.a. Explain two types of Bug algorithm and discuss how it enables a robot to navigate around obstacles to reach its target with the help of a diagram.(7.0 Marks)

4.b. Define temporal decomposition in the context of robotic navigation and illustrate it with a generic navigation architecture diagram. Briefly explain the four interrelated trends that correlate with temporal decomposition. (7.0 Marks)

4.c. Discuss any five key points of implementing Particle filter SLAM for localization and mapping in robotics? (7.0 Marks)

4.d. What are the advantages of using reinforcement learning in legged robots and autonomous vehicles? (4.0 Marks)